SIMATS ENGINEERING
ASSESSMENT TOOL 3 - UNIT 4
TECHNICAL PRESENTATION

COURSE NAME:PRINCIPLES OF DIGITAL SYSTEM DESIGN
COURSE CODE:ECA47

COURSE OUTCOME COVERED

CO4: Apply and evaluate microcontroller (8051/Arduino) architecture, instruction sets, and interfacing techniques to develop embedded system applications. (BL6)

Assessment Tool Weightage: 30%
Duration: 90 minutes
Marks: 100

This assessment is conducted as an individual or group presentation where students present a technical topic or solution based on course concepts. Students are required to explain the problem, methodology, analysis, and results using appropriate visuals such as slides or diagrams. The presentation should demonstrate clear understanding and effective communication. Evaluation is based on content accuracy, depth of knowledge, organization, presentation skills, and ability to answer questions.

TOPICS: Introduction to Microcontroller - Pin configuration - Architecture of 8051 - Instruction sets - Addressing modes-I/O Ports-External memory-Timers and Counters-Serial data-input and output-Interrupts- Interfacing with ADC, DAC, LEDs and seven segment display. Introduction to Arduino - Arduino IDE - Creating an Arduino program - Analog and Digital Interfacing

Criterion	Weightage (%)	Maximum Marks	Marks Awarded
Technical Content Accuracy	30%	**30**	______
Problem Identification	25%	**25**	______
Use of Tools	20%	**20**	______
Communication	15%	**15**	______
Professionalism	10%	**10**	______
Total	**100%**	**100**	**______ / 100**

QUESTIONS: [any 1 for a group of 4 students] 1 x 100 = 100

1. IoT Enabled Smart Smoke Detection and Monitoring system using MQ sensor and embedded technology
2. An IoT-based system for real-time monitoring and analysis of temperature and humidity with automated alerts
3. Designing of a smart electronic door, locking system using Aurdino and keypad technology
4. Implementation Of A Low-Cost Bluetooth-Based Smart Lighting System Using 8051
5. Predictive Smart Irrigation System for Efficient Water Conservation and Management
6. Smart Traffic Light Management system (based on Microcontroller)
7. Embedded Railway Track Integrity Monitoring System with Crack and Tilt Detection for Early Accident Prevention
8. Energy Efficient Smart Street Light Controller
9. Object detection and vibration alert to blind people
10. Footsteps generation analysis

Code of Conduct:

I ____________________ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Technical Content Accuracy	30	Highly accurate, in-depth, and insightful technical explanation	Technically correct content with adequate depth and clarity	Basic concepts presented with limited accuracy and depth	Content contains major technical errors; concepts poorly understood
Problem Identification	25	Problem rigorously analyzed using appropriate and advanced methods	Problem clearly defined with a logical engineering approach	Problem identified but analysis or methodology is weak	Problem statement unclear or incorrect; no logical approach
Use of Tools	20	Advanced tools, well-analyzed data, and highly effective visuals	Appropriate tools and data used effectively with clear visuals	Limited use of tools and basic data interpretation	Minimal or incorrect use of tools and data; poor visuals
Communication	15	Highly engaging, confident, and audience-focused delivery	Clear, organized, and professional communication	Understandable but lacks clarity, structure, or confidence	Poor organization, unclear speech, ineffective slides
Professionalism	10	Exemplary professionalism, leadership, and ethical awareness	Professional conduct with effective team collaboration	Basic professionalism; uneven team participation	Lacks professionalism; minimal contribution or ethical awareness